



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 11 YEARLY EXAMINATION

MATHEMATICS (2 UNIT)

Reading Time: 5 minutes
Length: two and half hours

Weighting: 40% towards final result

OUTCOMES Referred to: P1, P2, P3, P4, P5, P6, P7, P8

Instructions:

1. Attempt **ALL** questions
2. Show all necessary working
3. Each question is of equal value. Mark values are shown beside each part
4. **Begin** each question on a new sheet of paper. Ensure the *question number* is written at the beginning and your *name* and your *teacher's initials* (**PR/LP/AY/BL/DK/MOD/SW**) is written on the top of each sheet of paper
5. If you use more than one sheet of paper for a question, make sure you staple them together ensuring you have 10 bundles to hand in at the end
6. If you don't attempt a question, write your name on a piece of paper and the question number and **hand it in**
7. Board approved calculators are allowed and will be reset prior to entering the examination hall. *Students are reminded to reset their calculator back to Degrees Mode.*

Question 1 (10 marks)

- (a) Evaluate $\frac{4.3 \times 10^8}{5.6 \times 10^{12}}$ giving your answer in scientific notation correct to 2 significant figures

(1 mark)

- (b) Simplify: i. $\frac{3m}{3m+6}$

(1 mark)

ii. $\frac{y+7}{4} - \frac{y+3}{8}$

(2 marks)

- (c) Simplify: $\sqrt{125} - 5\sqrt{2} + 2\sqrt{5}$

(2 marks)

- (d) Solve: i. $2-3d=-7$

(1 mark)

ii. $|x-3|=5$

(2 marks)

- (e) Expand and simplify:
 $(3a-2b)^2$

(1 mark)

Question 2 (10 marks)

(a) Factorise completely:

i. $2a^2 - 6a$ (1 mark)

ii. $x^2 + 3x - 54$ (1 mark)

iii. $a^3 - a^2 + a - 1$ (2 marks)

iv. $8x^3 - 27$ (2 marks)

(b) Solve simultaneously:

$$\begin{aligned} x - y &= -2 \\ 2x + 3y &= 6 \end{aligned} \quad (2 \text{ marks})$$

(c) Solve: $x^2 - 4x > 0$ (2 marks)

Question 7 (10 marks)

- (a) Given $\sin \theta = \frac{5}{8}$ and $\cos \theta < 0$ find the exact value of $\cos \theta$ and $\tan \theta$
(2 marks)

- (b) Write down the exact value of:

i. $\cos 330^\circ$ (1 mark)

ii. $\sec 210^\circ$ (1 mark)

- (c) Simplify:

i. $\cot \theta \sin \theta$ (1 mark)

ii. $\frac{\cos \theta}{\operatorname{cosec} \theta - 1} + \frac{\cos \theta}{\operatorname{cosec} \theta + 1}$ (3 marks)

iii. $\frac{\sin \theta}{\cos \theta \tan \theta}$ (2 marks)

Question 8 (10 marks)

(a) If $f(x) = \begin{cases} -1 & \text{for } -3 \leq x < 0 \\ +1 & \text{for } x = 0 \\ 0 & \text{for } x > 0 \end{cases}$

Sketch this function

(1 mark)

(b) For the following graphs:

- i. Sketch on separate diagrams showing their main features
- ii. State the domain and range

$\alpha \quad y = |x - 1|$ (3 marks)

$\beta \quad x^2 + y^2 = 9$ (2 marks)

$\chi \quad y = 2^x$ (3 marks)

(c) State the largest possible domain for the function $f(x) = \sqrt{4 - x^2}$
(1 mark)

Question 9 (10 marks)

(a) Simplify: $\frac{3^n \times 9^{n+1}}{27^n}$ (2 marks)

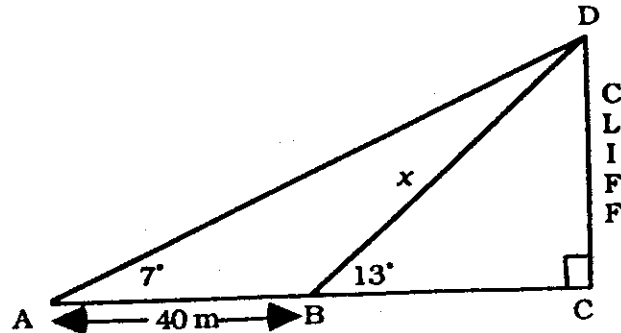
(b) Given that $5x^2 - 2x + 3 \equiv A(x-1) + B(x-1) + C$ for all values of x ,
find the value of A , B and C (2 marks)

(c) Find the centre and radius of the circle with equation
 $x^2 - 2x + y^2 + 6y - 6 = 0$ (4 marks)

(d) Simplify: $\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}$ (2 marks)

Question 10 (10 marks)

- (a) The angle of elevation from B to the top of a cliff is 13° . Katie walks 40 metres away from B and notes that the angle of elevation is now 7° .



- i. Find the value of x to 2 decimal places (2 marks)
 - ii. Find the height of the cliff to the nearest metre (2 marks)
- (b) Simplify: $5 - 3\cos^2 \theta - 3\sin^2 \theta$ (2 marks)
- (c) If α and β are the roots of the equation $2x^2 + 6x + 3 = 0$, find the value of:
- i. $\alpha + \beta$ (1 mark)
 - ii. $\alpha\beta$ (1 mark)
 - iii. $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (2 marks)

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Yr 11 Yearly Solutions 2007

1. a) $7.6785 \times 10^{-5} = 7.7 \times 10^{-5}$ ①

b) i. $\frac{3m}{3(m+2)} = \frac{m}{m+2}$ ①

ii. $\frac{y+7}{4} - \frac{y+3}{8} = \frac{2y+14-y-3}{8}$ ①

$= \frac{y+11}{8}$ ①

c) $\sqrt{125} - 5\sqrt{2} + 2\sqrt{5} = 5\sqrt{5} - 5\sqrt{2} + 2\sqrt{5}$ ① for $5\sqrt{5}$
 $= 7\sqrt{5} - 5\sqrt{2}$ ①

d) i. $2 - 3d = -7$
 $-3d = -9$
 $d = 3$ ①

ii. $|x-3| = 5$
 $x-3=5$
 $x=8$ ①

$-x+3=5$
 $-x=2$
 $x=-2$ ①

e) $(3a-2b)^2 = 9a^2 - 12ab + 4b^2$ ①

$$2. a) i. 2a^2 - 6a = 2a(a-3) \quad (1)$$

$$ii. x^2 + 3x - 54 = (x+9)(x-6) \quad (1)$$

$$iii. a^3 - a^2 + a - 1 = a^2(a-1) + (a-1) \quad (1)$$

$$= (a-1)(a^2+1) \quad (1)$$

$$iv. 8x^3 - 27 = (2x-3)(4x^2 + 6x + 9)$$

$$b) \quad \begin{aligned} x - y &= -2 \quad \dots 1 \\ 2x + 3y &= 6 \quad \dots 2 \end{aligned}$$

$$\begin{aligned} x &= y - 2 \\ \text{Subs } x &= y - 2 \text{ into } \dots 2 \end{aligned}$$

$$\begin{aligned} 2(y-2) + 3y &= 6 \\ 2y - 4 + 3y &= 6 \\ 5y &= 10 \\ y &= 2 \end{aligned}$$

1 for method
1 for answer

$$\begin{aligned} x &= 2 - 2 \\ &= 0 \end{aligned} \quad \therefore (0, 2)$$

$$c) \quad \begin{aligned} x^2 - 4x &> 0 \\ x(x-4) &> 0 \end{aligned}$$

1 for factorising

$$x < 0, \quad x > 4$$

1 for both answers

3. a) $\angle YRV = \angle XTV = 90^\circ$ (in rectangle)
 $\angle RYV = \angle YVT$ ($RY \parallel VT$, alternate $\angle$ s)
 $\angle VXT = \angle YVR$ ($VR \parallel TS$, alternate $\angle$ s)

① For each reasoning

$\therefore \triangle RYV \parallel \triangle TVX$ (equiangular)

① For 'equiangular' reason

4 total

b) $f(x) = x^2 - 5x + 6$

① For correct substitution

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - 5(x+h) + 6 - (x^2 - 5x + 6)}{h}$$

1 mark $\left\{ \begin{aligned} &= \frac{x^2 + 2xh + h^2 - 5x - 5h + 6 - x^2 + 5x - 6}{h} \\ &= \frac{2xh + h^2 - 5h}{h} \end{aligned} \right.$

$= 2x + h - 5$

① For answer

c) $2x^2 + (1-k)x + k-3 = 0$

roots are $\alpha, 2\alpha$

$$\alpha + 2\alpha = \frac{-b}{a} = \frac{-(1-k)}{2}$$

$$3\alpha = \frac{-1+k}{2}$$

$$\alpha = \frac{-1+k}{6}$$

1 mark for $\alpha, 2\alpha$
 1 mark for same method
 1 mark for answer

$$\alpha \times 2\alpha = \frac{c}{a}$$

$$2\alpha^2 = \frac{k-3}{2}$$

$$2\left(\frac{k-1}{6}\right)^2 = \frac{k-3}{2}$$

$$\frac{k^2 - 2k + 1}{36} = \frac{k-3}{2}$$

$$4k^2 - 8k + 4 = 36k - 108$$

$$4k^2 - 44k + 112 = 0$$

$$k^2 - 11k + 28 = 0$$

$(k-7)(k-4) = 0$

$\therefore k = 4, 7$

4. a) i. $y = 4x^3 - x^2 + 8$
 $y' = 12x^2 - 2x$

①

ii. $y = (x^2 - 1)^{\frac{1}{2}}$

$y' = \frac{1}{2} (x^2 - 1)^{-\frac{1}{2}} \cdot 2x$

①

$= \frac{x}{\sqrt{x^2 - 1}}$

①

iii. $y = \frac{x^2 + x}{x}$

$= x + 1$

$y' = 1$

①
①

iv. $y = \frac{x+5}{x+2}$

$u = x+5$

$u' = 1$

$v = x+2$

$v' = 1$

$y' = \frac{vu' - uv'}{v^2}$

$= \frac{x+2 - (x+5)}{(x+2)^2}$

$= \frac{x+2 - x - 5}{(x+2)^2}$

$= \frac{-3}{(x+2)^2}$

1 for method
1 for answer

v. $y = 2x^4 (x+1)^3$

$u = 2x^4$ $v = (x+1)^3$

$u' = 8x^3$ $v' = 3(x+1)^2$

1 method

1 subs

1 simplified answer

$y' = 8x^3 (x+1)^3 +$

$2x^4 \times 3(x+1)^2$
 $= 2x^3 (x+1)^2 [4(x+1) + 3x]$
 $= 2x^3 (x+1)^2 (7x+4)$

5. a) $\frac{a^8 b^{-12}}{a^3 b^{-2}} = a^5 b^{-10}$
 $= \frac{a^5}{b^{10}}$ } 1 for either answer

b) $x = 4.23434...$
 $10x = 42.3434...$
 $100x = 423.434...$
 $1000x = 4234.34...$

1 method

990 $x = 4192$
 $x = \frac{4192}{990}$
 $= 4 \frac{116}{495}$

1 answer

c) $y = 3x^2 + 2$
 $y' = 6x$
 $f'(1) = 6$

when $x=1$, $y=5$

1 for y value
 1 for m
 1 for eqn

$y - y_1 = m(x - x_1)$
 $y - 5 = 6(x - 1)$
 $y - 5 = 6x - 6$
 $y = 6x - 1$

d) α . i. B, C
 ii. between B and C

β . $0 < \text{grad} < 7$

2 if full range
 1 if only integers given

6 a)

b)

$$M = \left(\frac{-1+1}{2}, \frac{0+6}{2} \right)$$

$$= (0, 3)$$

1 for substitution

1 for answer

c)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0-3}{-1-0}$$

$$= 3$$

$$y - 3 = 3(x - 0)$$

$$y = 3x + 3$$

d)

$$3x + y - 9 = 0$$

$$y = 0$$

$$3x = 9$$

$$x = 3$$

$$\therefore (3, 0)$$

e)

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

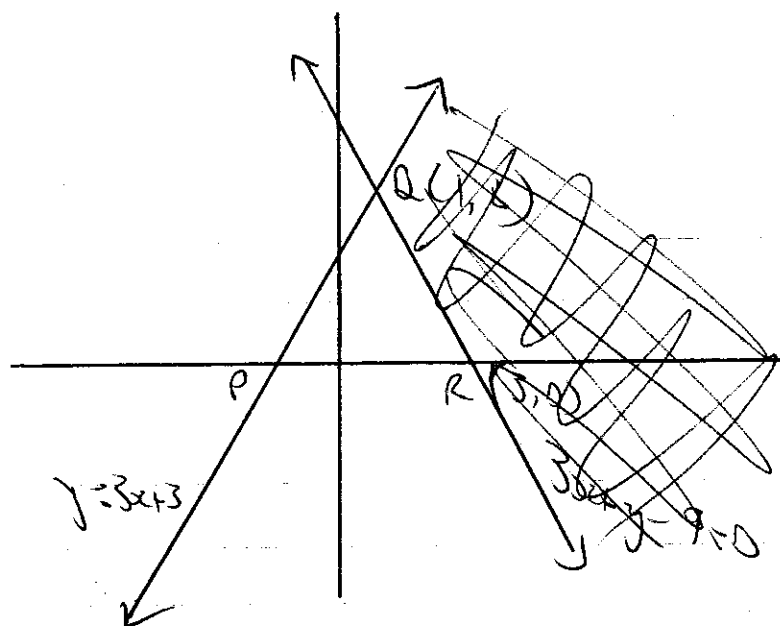
$$= \frac{|3x - 1 + 1 \cdot 0 - 9|}{\sqrt{3^2 + 1^2}}$$

$$= \frac{|-12|}{\sqrt{10}}$$

$$= \frac{12}{\sqrt{10}}$$

$$= \frac{12}{\sqrt{10}}$$

1 for substitute



$$y \leq 3x + 3$$

$$3x + y - 9 > 0$$

Test (0, 0)

$$0 \leq 3 \quad \checkmark$$

$$-9 > 0 \quad \times$$

Test (4, 0)

$$0 \leq 15 \quad \checkmark$$

$$3(4) - 9 > 0$$

$$3 > 0 \quad \checkmark$$

1 for shading
1 for testing

7. a)



$$\cos \theta = -\frac{\sqrt{39}}{8}$$

$$\tan \theta = -\frac{5}{\sqrt{39}}$$

1 for sign
1 for values

b) i. $\cos 330 = \cos 30$
 $= \frac{\sqrt{3}}{2}$ ①

ii. $\sec 210 = \frac{1}{\cos 210}$
 $= \frac{1}{-\cos 30}$
 $= -\frac{2}{\sqrt{3}}$ ①

c) i. $\cot \theta \sin \theta = \frac{\cos \theta}{\cancel{\sin \theta}} \times \cancel{\sin \theta}$
 $= \cos \theta$ ①

ii. $\frac{\cos \theta}{\operatorname{cosec} \theta - 1} + \frac{\cos \theta}{\operatorname{cosec} \theta + 1} = \frac{\cos \theta (\operatorname{cosec} \theta + 1) + \cos \theta (\operatorname{cosec} \theta - 1)}{(\operatorname{cosec}^2 \theta - 1)}$
 $= \frac{\cot \theta + \cos \theta + \cot \theta - \cos \theta}{\operatorname{cosec}^2 \theta - 1}$
 $= \frac{2 \cot \theta}{\operatorname{cosec}^2 \theta - 1}$ ①
 $= \frac{2 \cot \theta}{\cot^2 \theta}$
 $= \frac{2}{\cot \theta}$
 $= 2 \tan \theta$ or $\frac{2}{\cot \theta}$ ①

① for fraction

iii.

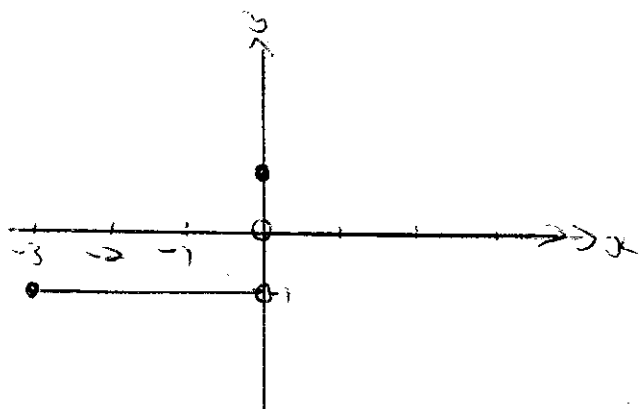
$$\frac{\sin \theta}{\cos \theta \tan \theta} = \frac{\sin \theta}{\cancel{\cos \theta} \times \frac{\sin \theta}{\cancel{\cos \theta}}}$$

$$= \frac{\sin \theta}{\sin \theta}$$

$$= 1$$

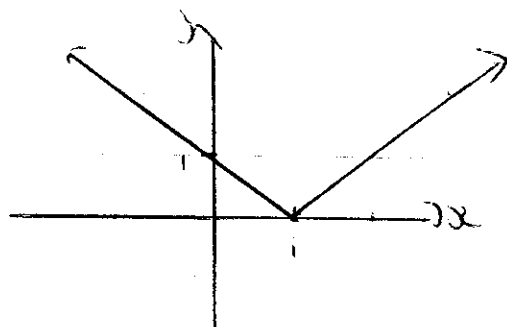
} ①
①

8. a)



①

b) 2. $y = |x - 1|$

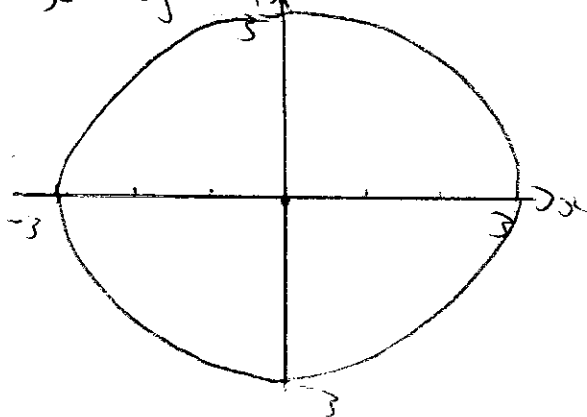


D: all real values of x
R: $y \geq 0$

1 mark for graph
1 each for D & R

3 total

B. $x^2 + y^2 = 9$



D: $-3 \leq x \leq 3$

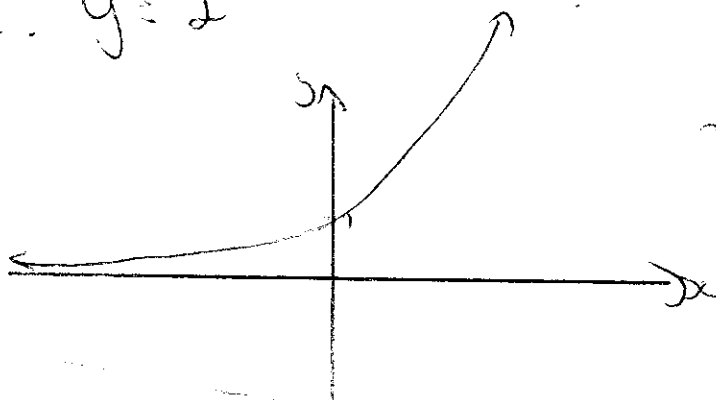
R: $-3 \leq y \leq 3$

1 mark for diag

1 mark for D & R

2 total

X. $y = 2^x$



D: all real values of x
R: $y > 0$

1 mark for graph
1 mark each for D & R
3 total

c) D: $-2 \leq x \leq 2$ ①

$$\begin{aligned}
 9. a) \quad \frac{3^n \times 9^{n+1}}{27^n} &= \frac{3^n \times (3^2)^{n+1}}{3^{3n}} \\
 &= \frac{3^n \times 3^{2n+2}}{3^{3n}} \\
 &= \frac{3^{3n+2}}{3^{3n}} \\
 &= 3^2 \\
 &= 9
 \end{aligned}$$

1 for method

1 mark

$$b) 5x^2 - 2x + 3 \equiv A(x-1)^2 + B(x-1) + C$$

$$\begin{aligned}
 \text{RHS: } A(x^2 - 2x + 1) + Bx - B + C \\
 &= Ax^2 - 2Ax + A + Bx - B + C \\
 &= Ax^2 + x(B - 2A) + A - B + C
 \end{aligned}$$

$$\therefore A = 5$$

$$B - 2A = -2$$

$$A - B + C = 3$$

$$B - 2A = -2$$

$$B - 2(5) = -2$$

$$B = 8$$

$$A - B + C = 3$$

$$5 - 8 + C = 3$$

$$C = 6$$

2 total

$$c) x^2 - 2x + y^2 + 6y - 6 = 0$$

$$x^2 - 2x + 1 + y^2 + 6y + 9 = 6 + 1 + 9 \quad ①$$

$$(x-1)^2 + (y+3)^2 = 16 \quad ①$$

$$\text{centre: } (1, -3)$$

$$\text{radius: } 4$$

①
①
①

4 total

$$d) \left. \begin{array}{l} \frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}} \quad 1 + \frac{1}{\sqrt{2}} \div \left(1 - \frac{1}{\sqrt{2}}\right) \\ \frac{\sqrt{2} + 1}{\sqrt{2}} \div \left(\frac{\sqrt{2} - 1}{\sqrt{2}}\right) \end{array} \right\} \textcircled{1}$$

$$\frac{\sqrt{2} + 1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2} - 1} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \quad \textcircled{1}$$

total 2

10. a) i. $\angle DBA = 167^\circ$
 $\therefore \angle BDA = 6^\circ$

$$\frac{40}{\sin 6} = \frac{x}{\sin 7}$$

① for substitution

$$x \sin 6 = 40 \sin 7$$

$$x = \frac{40 \sin 7}{\sin 6}$$

$$= 46.63585 \dots$$

$$= 46.64 \text{ m}$$

rounding error

① for answer

ii. $\sin B = \frac{\text{diff}}{46.64}$

①

$$\text{diff} = 10.49 \dots$$

$$= 10 \text{ m}$$

①

b) $5 - 3 \cos^2 \theta - 3 \sin^2 \theta$

$$= 5 - 3 \cos^2 \theta - 3(1 - \cos^2 \theta)$$

$$= 5 - 3 \cos^2 \theta - 3 + 3 \cos^2 \theta$$

$$= 2$$

① for trig identity

①

c) $2x^2 + 6x + 3 = 0$

i. $\alpha + \beta = \frac{-b}{a}$

$$= \frac{-6}{2}$$

$$= -3$$

①

ii. $\alpha \beta = \frac{c}{a}$

$$= \frac{3}{2}$$

①

iii. $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha \beta}$

①

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(-3)^2 - 2\left(\frac{3}{2}\right)}{\frac{3}{2}} = 4$$

①